Final Year Research Project Proposal

Project Title:
CogniCare+: A Smart, AI-Powered System for Early Detection, Monitoring, and Support for Dementia Patients

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1. Introduction:
Dementia affects over 55 million people worldwide, with nearly 10 million new cases every year. Early detection and continuous care are critical in managing dementia and improving patient outcomes. Our project proposes the development of a smart, AI-powered multi-component system that detects early signs of dementia, monitors daily activity, predicts disease progression, and supports caregivers with actionable insights. The system leverages AI, IoT, NLP, time-series forecasting, and knowledge graphs.

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2. Research Problem Statement:
Despite advances in medical diagnostics, many dementia cases are detected too late, and there is minimal integration of AI-driven assistance in daily care. How can we develop an intelligent, interpretable, and personalized system that detects early dementia, tracks progression, and supports real-time decision-making for caregivers?

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3. Project Objectives:

- To build AI models capable of early detection of dementia using speech and writing behavior.
- To develop a real-time activity monitoring system using smart sensors for anomaly detection.
- To implement predictive models for dementia progression using longitudinal data and explainable AI.
- To create a knowledge-based decision support system for caregivers with actionable alerts and reports.

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4. Team Member Scopes:

✉️ **Member 1 – Behavioral and Linguistic Analysis (ML/NLP Focus)**
Scope Title: *AI-Driven Early Detection of Dementia Using Behavioral and Linguistic Cues*
Focus Areas:

- Train and evaluate models (SVM, CNN, Transformers) on speech and text datasets (DementiaBank, ADReSS).
- Analyze sentence structure, lexical variety, and hesitation patterns.
- Compare traditional ML vs. transformer-based models (e.g., BERT).
- Use transfer learning from general NLP models to fine-tune dementia-specific predictors.

Research Component:

- Early symptom identification through language biomarkers.
- Contribution to cognitive AI and clinical NLP.

Technologies: Python, scikit-learn, BERT, HuggingFace, Jupyter Notebooks.

🔹 **Member 2 – Activity Recognition via IoT and Smart Sensors**

Scope Title: *Sensor-Based Real-Time Monitoring and Anomaly Detection in Dementia Patients*
Focus Areas:

- Develop edge-device compatible activity recognition using sensor data.
- Classify daily activities and detect abnormal patterns (e.g., wandering, prolonged inactivity).
- Compare edge (e.g., Raspberry Pi) vs. cloud inference performance.
- Integrate reminders, alarms, and to-do list notifications.

Research Component:

- Behavior modeling, time-series anomaly detection.
- IoT performance optimization.

Technologies: Arduino, Raspberry Pi, Python, MQTT, Node-RED.

🔹 **Member 3 – Predictive Modeling & Explainable AI**

Scope Title: *Explainable Predictive Modeling of Dementia Progression Using Longitudinal Behavioral and Clinical Data*
Focus Areas:

- Build LSTM, Transformer, and GRU-based models to forecast stage transitions.
- Use clinical and behavioral time-series data (ADNI, OASIS-3).
- Implement SHAP, LIME, and attention visualization.
- Evaluate model trustworthiness and interpretability.

Research Component:

- Time-series modeling for cognitive health forecasting.
- Explainability for ethical and clinical AI.

Technologies: PyTorch, TensorFlow, SHAP, LIME, Matplotlib, Dash.

🔹 **Member 4 – AI-Powered Decision Support System**

Scope Title: *Intelligent Decision Support for Dementia Caregivers Using Context-Aware AI*
Focus Areas:

- Design knowledge graph for patient status, history, medication, and behaviors.
- Create rule-based and ML-enhanced reasoning agents.
- Build a dashboard with medical reports, consultation tracking, reminders, and care recommendations.
- Integrate alerts and notification system.

Research Component:

- Hybrid AI reasoning, semantic healthcare systems.
- Usability and effectiveness of intelligent care assistants.

Technologies: Neo4j, Prolog (optional), React/Flask, REST APIs, Firebase.

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5. Methodology Summary:

- Literature Review on AI in Dementia Care.
- Dataset collection and preprocessing.
- Module-wise development (NLP, IoT, Forecasting, Knowledge System).
- Integration and Testing.
- Clinical validation (simulated) and caregiver feedback.

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6. Expected Outcomes:

- Functional prototype combining detection, prediction, and support tools.
- Research contributions in AI-for-healthcare, NLP in cognitive detection, and XAI.
- Publishable results in student conferences or AI/HealthTech symposiums.

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7. Timeline:

- Month 1–2: Research, Dataset Acquisition, Planning
- Month 3–5: Module Development (Parallel Work)
- Month 6: Integration and Internal Testing
- Month 7: Evaluation, Final Report, and Submission

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8. Tools and Resources Required:

- Access to datasets (DementiaBank, ADNI, etc.)
- IoT hardware (Raspberry Pi, sensors, camera, microphone)
- GPUs for model training (Colab, local GPUs)

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9. Risk and Mitigation:

- **Data Privacy Risks:** Use synthetic datasets or anonymized samples.
- **Hardware Limitations:** Simulate sensors if real-time access fails.
- **Complexity of Integration:** Use modular testing and containerization.

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10. Supervisor Expectations:

- Guide literature review and ethical compliance
- Validate methodology and evaluation metrics
- Support in aligning scope with academic research standards